



Importing Libraries

```
In [264... import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
plt.style.use('dark_background')
import plotly.express as px
import plotly.io as pio
pio.templates.default = "plotly_dark"
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

Reading CSV

```
In [265... df = pd.read_csv('/content/zomato.csv', engine='python', on_bad_lines='skip')
df.head()
```

```
Out[265...
```

	url	address	name	online_order	book_table	ra
0	https://www.zomato.com/bangalore/jalsa-banasha...	942, 21st Main Road, 2nd Stage, Banashankari, ...	Jalsa	Yes	Yes	4.1
1	https://www.zomato.com/bangalore/spice-elephan...	2nd Floor, 80 Feet Road, Near Big Bazaar, 6th ...	Spice Elephant	Yes	No	4.1
2	https://www.zomato.com/SanchurroBangalore?cont...	1112, Next to KIMS Medical College, 17th Cross...	San Churro Cafe	Yes	No	3.8
3	https://www.zomato.com/bangalore/addhuri-udupi...	1st Floor, Annakuteera, 3rd Stage, Banashankar...	Addhuri Udupi Bhojana	No	No	3.7
4	https://www.zomato.com/bangalore/grand-village...	10, 3rd Floor, Lakshmi Associates, Gandhi Baza...	Grand Village	No	No	3.8

```
In [266... df.shape
```

Out[266... (51155, 17)

```
In [267... df.columns
```

Out[267... Index(['url', 'address', 'name', 'online_order', 'book_table', 'rate', 'votes',
 'phone', 'location', 'rest_type', 'dish_liked', 'cuisines',
 'approx_cost(for two people)', 'reviews_list', 'menu_item',
 'listed_in(type)', 'listed_in(city)'],
 dtype='object')

```
In [268... df = df.drop(['url', 'address', 'phone', 'menu_item', 'dish_liked', 'reviews_l  
df.head()
```

Out[268...

	name	online_order	book_table	rate	votes	location	rest_type	cui
0	Jalsa	Yes	Yes	4.1/5	775	Banashankari	Casual Dining	Ir Mu Ch
1	Spice Elephant	Yes	No	4.1/5	787	Banashankari	Casual Dining	Chi Ir
2	San Churro Cafe	Yes	No	3.8/5	918	Banashankari	Cafe, Casual Dining	Me: I
3	Addhuri Udupi Bhojana	No	No	3.7/5	88	Banashankari	Quick Bites	Ir I
4	Grand Village	No	No	3.8/5	166	Basavanagudi	Casual Dining	Ir Rajas

```
In [269... df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 51155 entries, 0 to 51154
Data columns (total 11 columns):
#   Column                                          Non-Null Count  Dtype
---  -
0   name                                           51155 non-null  object
1   online_order                                  51155 non-null  object
2   book_table                                    51155 non-null  object
3   rate                                           43380 non-null  object
4   votes                                          51155 non-null  int64
5   location                                      51134 non-null  object
6   rest_type                                    50931 non-null  object
7   cuisines                                      51110 non-null  object
8   approx_cost(for two people)                 50811 non-null  object
9   listed_in(type)                             51155 non-null  object
10  listed_in(city)                             51155 non-null  object
dtypes: int64(1), object(10)
memory usage: 4.3+ MB

```

Dropping Duplicates

```

In [270...] df.drop_duplicates(inplace = True)
df.shape

```

Out[270...] (51049, 11)

Cleaning Rate Column

```

In [271...] df['rate'].unique()

```

```

Out[271...] array(['4.1/5', '3.8/5', '3.7/5', '3.6/5', '4.6/5', '4.0/5', '4.2/5',
        '3.9/5', '3.1/5', '3.0/5', '3.2/5', '3.3/5', '2.8/5', '4.4/5',
        '4.3/5', 'NEW', '2.9/5', '3.5/5', nan, '2.6/5', '3.8 /5', '3.4/5',
        '4.5/5', '2.5/5', '2.7/5', '4.7/5', '2.4/5', '2.2/5', '2.3/5',
        '3.4 /5', '-', '3.6 /5', '4.8/5', '3.9 /5', '4.2 /5', '4.0 /5',
        '4.1 /5', '3.7 /5', '3.1 /5', '2.9 /5', '3.3 /5', '2.8 /5',
        '3.5 /5', '2.7 /5', '2.5 /5', '3.2 /5', '2.6 /5', '4.5 /5',
        '4.3 /5', '4.4 /5', '4.9/5', '2.1/5', '2.0/5', '1.8/5', '4.6 /5',
        '4.9 /5', '3.0 /5', '4.8 /5', '2.3 /5', '2.4 /5', '4.7 /5',
        '2.2 /5', '2.0 /5', '2.1 /5', '1.8 /5'], dtype=object)

```

Removing "NEW" , "-" and "/5" from Rate Column

```

In [272...] def handlerate(value):
    if(value=='NEW' or value=='-'):
        return np.nan
    else:
        value = str(value).split('/')

```

```

        value = value[0]
        return float(value)

df['rate'] = df['rate'].apply(handlerate)
df['rate'].head()

```

Out[272...

	rate
0	4.1
1	4.1
2	3.8
3	3.7
4	3.8

dtype: float64

Filling Null Values in Rate Column with Mean

```

In [273... df['rate'].fillna(df['rate'].mean(), inplace = True)
df['rate'].isnull().sum()

```

Out[273... np.int64(0)

In [274... df.info()

```

<class 'pandas.core.frame.DataFrame'>
Index: 51049 entries, 0 to 51154
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   name                                  51049 non-null  object
1   online_order                          51049 non-null  object
2   book_table                            51049 non-null  object
3   rate                                  51049 non-null  float64
4   votes                                51049 non-null  int64
5   location                              51028 non-null  object
6   rest_type                             50825 non-null  object
7   cuisines                              51004 non-null  object
8   approx_cost(for two people)           50707 non-null  object
9   listed_in(type)                       51049 non-null  object
10  listed_in(city)                       51049 non-null  object
dtypes: float64(1), int64(1), object(9)
memory usage: 4.7+ MB

```

Dropping Null Values

```

In [275... df.dropna(inplace = True)

```

```
df.head()
```

Out[275...

	name	online_order	book_table	rate	votes	location	rest_type	cuis
0	Jalsa	Yes	Yes	4.1	775	Banashankari	Casual Dining	I In Mug Ch
1	Spice Elephant	Yes	No	4.1	787	Banashankari	Casual Dining	Chi I In
2	San Churro Cafe	Yes	No	3.8	918	Banashankari	Cafe, Casual Dining	Mex It
3	Addhuri Udupi Bhojana	No	No	3.7	88	Banashankari	Quick Bites	S In I It
4	Grand Village	No	No	3.8	166	Basavanagudi	Casual Dining	I In Rajas

In [276...

```
df.rename(columns = {'approx_cost(for two people)':'Cost2plates', 'listed_in(t  
df.head()
```

Out[276...

	name	online_order	book_table	rate	votes	location	rest_type	cuis
0	Jalsa	Yes	Yes	4.1	775	Banashankari	Casual Dining	I In Mug Ch
1	Spice Elephant	Yes	No	4.1	787	Banashankari	Casual Dining	Chi I In
2	San Churro Cafe	Yes	No	3.8	918	Banashankari	Cafe, Casual Dining	Mex It
3	Addhuri Udupi Bhojana	No	No	3.7	88	Banashankari	Quick Bites	S In I It
4	Grand Village	No	No	3.8	166	Basavanagudi	Casual Dining	I In Rajas

In [277...

```
df['location'].unique()
```

```
Out[277... array(['Banashankari', 'Basavanagudi', 'Mysore Road', 'Jayanagar',
      'Kumaraswamy Layout', 'Rajarajeshwari Nagar', 'Vijay Nagar',
      'Uttarahalli', 'JP Nagar', 'South Bangalore', 'City Market',
      'Nagarbhavi', 'Bannerghatta Road', 'BTM', 'Kanakapura Road',
      'Bommanahalli', 'CV Raman Nagar', 'Electronic City', 'HSR',
      'Marathahalli', 'Wilson Garden', 'Shanti Nagar',
      'Koramangala 5th Block', 'Koramangala 8th Block', 'Richmond Road',
      'Koramangala 7th Block', 'Jalahalli', 'Koramangala 4th Block',
      'Bellandur', 'Sarjapur Road', 'Whitefield', 'East Bangalore',
      'Old Airport Road', 'Indiranagar', 'Koramangala 1st Block',
      'Frazer Town', 'RT Nagar', 'MG Road', 'Brigade Road',
      'Lavelle Road', 'Church Street', 'Ulsoor', 'Residency Road',
      'Shivajinagar', 'Infantry Road', 'St. Marks Road',
      'Cunningham Road', 'Race Course Road', 'Commercial Street',
      'Vasanth Nagar', 'HBR Layout', 'Domlur', 'Ejipura',
      'Jeevan Bhima Nagar', 'Old Madras Road', 'Malleshwaram',
      'Seshadripuram', 'Kammanahalli', 'Koramangala 6th Block',
      'Majestic', 'Langford Town', 'Central Bangalore', 'Sanjay Nagar',
      'Brookefield', 'ITPL Main Road, Whitefield',
      'Varthur Main Road, Whitefield', 'KR Puram',
      'Koramangala 2nd Block', 'Koramangala 3rd Block', 'Koramangala',
      'Hosur Road', 'Rajajinagar', 'Banaswadi', 'North Bangalore',
      'Nagawara', 'Hennur', 'Kalyan Nagar', 'New BEL Road', 'Jakkur',
      'Rammurthy Nagar', 'Thippasandra', 'Kaggadasapura', 'Hebbal',
      'Kengeri', 'Sankey Road', 'Sadashiv Nagar', 'Basaveshwara Nagar',
      'Yeshwantpur', 'West Bangalore', 'Magadi Road', 'Yelahanka',
      'Sahakara Nagar', 'Peenya'], dtype=object)
```

```
In [278... df['listed_in(city)'].unique()
```

```
Out[278... array(['Banashankari', 'Bannerghatta Road', 'Basavanagudi', 'Bellandur',
      'Brigade Road', 'Brookefield', 'BTM', 'Church Street',
      'Electronic City', 'Frazer Town', 'HSR', 'Indiranagar',
      'Jayanagar', 'JP Nagar', 'Kalyan Nagar', 'Kammanahalli',
      'Koramangala 4th Block', 'Koramangala 5th Block',
      'Koramangala 6th Block', 'Koramangala 7th Block', 'Lavelle Road',
      'Malleshwaram', 'Marathahalli', 'MG Road', 'New BEL Road',
      'Old Airport Road', 'Rajajinagar', 'Residency Road',
      'Sarjapur Road', 'Whitefield'], dtype=object)
```

Listed in(city) and location, both are there, lets keep only one.

```
In [279... df = df.drop(['listed_in(city)'], axis = 1)
```

```
In [280... df['Cost2plates'].unique()
```

```
Out[280...] array(['800', '300', '600', '700', '550', '500', '450', '650', '400',  
                  '900', '200', '750', '150', '850', '100', '1,200', '350', '250',  
                  '950', '1,000', '1,500', '1,300', '199', '80', '1,100', '160',  
                  '1,600', '230', '130', '50', '190', '1,700', '1,400', '180',  
                  '1,350', '2,200', '2,000', '1,800', '1,900', '330', '2,500',  
                  '2,100', '3,000', '2,800', '3,400', '40', '1,250', '3,500',  
                  '4,000', '2,400', '2,600', '120', '1,450', '469', '70', '3,200',  
                  '60', '560', '240', '360', '6,000', '1,050', '2,300', '4,100',  
                  '5,000', '3,700', '1,650', '2,700', '4,500', '140'], dtype=object)
```

Removing , from Cost2Plates Column

```
In [281...] def handlecomma(value):  
             value = str(value)  
             if ',' in value:  
                 value = value.replace(',', '')  
                 return float(value)  
             else:  
                 return float(value)  
  
df['Cost2plates'] = df['Cost2plates'].apply(handlecomma)  
df['Cost2plates'].unique()
```

```
Out[281...] array([ 800.,  300.,  600.,  700.,  550.,  500.,  450.,  650.,  400.,  
                  900.,  200.,  750.,  150.,  850.,  100., 1200.,  350.,  250.,  
                  950., 1000., 1500., 1300.,  199.,   80., 1100.,  160., 1600.,  
                  230.,  130.,   50.,  190., 1700., 1400.,  180., 1350., 2200.,  
                 2000., 1800., 1900.,  330., 2500., 2100., 3000., 2800., 3400.,  
                   40., 1250., 3500., 4000., 2400., 2600.,  120., 1450.,  469.,  
                   70., 3200.,   60.,  560.,  240.,  360., 6000., 1050., 2300.,  
                 4100., 5000., 3700., 1650., 2700., 4500.,  140.]
```

```
In [282...] df.head()
```

Out[282...

	name	online_order	book_table	rate	votes	location	rest_type	cuis
0	Jalsa	Yes	Yes	4.1	775	Banashankari	Casual Dining	I In Muq Ch
1	Spice Elephant	Yes	No	4.1	787	Banashankari	Casual Dining	Chi I In
2	San Churro Cafe	Yes	No	3.8	918	Banashankari	Cafe, Casual Dining	Mex It
3	Addhuri Udupi Bhojana	No	No	3.7	88	Banashankari	Quick Bites	S In I Ir
4	Grand Village	No	No	3.8	166	Basavanagudi	Casual Dining	I In Rajas

Cleaning Rest Type Column

In [283...

```
rest_types = df['rest_type'].value_counts(ascending = False)
rest_types
```


Out[283...

	count
rest_type	
Quick Bites	18979
Casual Dining	10135
Cafe	3621
Delivery	2563
Dessert Parlor	2232
...	...
Dessert Parlor, Kiosk	2
Dessert Parlor, Food Court	2
Food Court, Beverage Shop	2
Sweet Shop, Dessert Parlor	1
Quick Bites, Kiosk	1

93 rows × 1 columns

dtype: int64

In [284...

```
rest_types_lessthan1000 = rest_types[rest_types<1000]
rest_types_lessthan1000
```

Out[284...

rest_type	count
Beverage Shop	863
Bar	668
Food Court	616
Sweet Shop	468
Bar, Casual Dining	388
...	...
Dessert Parlor, Kiosk	2
Dessert Parlor, Food Court	2
Food Court, Beverage Shop	2
Sweet Shop, Dessert Parlor	1
Quick Bites, Kiosk	1

85 rows × 1 columns

dtype: int64

Making Rest Types less than 1000 in frequency as others

In [285...

```
def handle_rest_type(value):
    if (value in rest_types_less_than_1000):
        return 'others'
    else:
        return value

df['rest_type'] = df['rest_type'].apply(handle_rest_type)
df['rest_type'].value_counts()
```

Out[285...

	count
rest_type	
Quick Bites	18979
Casual Dining	10135
others	8737
Cafe	3621
Delivery	2563
Dessert Parlor	2232
Takeaway, Delivery	2005
Bakery	1139
Casual Dining, Bar	1076

dtype: int64

Cleaning Location Column

```
In [286... location = df['location'].value_counts(ascending = False)

location_lessthan300 = location[location < 300]

df['location'] = df['location'].apply(
    lambda x: 'others' if x in location_lessthan300 else x
)
```

Cleaning Cuisines Column

```
In [287... cuisines = df['cuisines'].value_counts(ascending = False)

cuisines_lessthan100 = cuisines[cuisines<100]

def handle_cuisines(value):
    if(value in cuisines_lessthan100):
        return 'others'
    else:
        return value

df['cuisines'] = df['cuisines'].apply(handle_cuisines)
df['cuisines'].value_counts()
```

Out[287...

count	
cuisines	
others	25883
North Indian	2829
North Indian, Chinese	2337
South Indian	1817
Biryani	902
...	...
Chinese, Thai, Momos	111
Cafe, Bakery	109
South Indian, Chinese, North Indian	105
South Indian, Fast Food	104
North Indian, Chinese, Seafood	102

68 rows × 1 columns

dtype: int64

In [288...

```
df.head()
```

Out[288...

	name	online_order	book_table	rate	votes	location	rest_type	cuisi
0	Jalsa	Yes	Yes	4.1	775	Banashankari	Casual Dining	otr
1	Spice Elephant	Yes	No	4.1	787	Banashankari	Casual Dining	otr
2	San Churro Cafe	Yes	No	3.8	918	Banashankari	others	otr
3	Addhuri Udupi Bhojana	No	No	3.7	88	Banashankari	Quick Bites	Sc Ind No Inc
4	Grand Village	No	No	3.8	166	Basavanagudi	Casual Dining	otr

Data Cleaning Done!, Lets jump to Visualization.

Count Plot of Various Locations

```
In [289... fig = px.histogram(df, y='location', color='online_order', orientation='h',  
  
    labels={  
        'location': 'Restaurant Location',  
        'online_order': 'Online Order Available'  
    },  
  
    color_discrete_map={'Yes': '#00CC96', 'No': '#EF553B'},  
  
    title='Online Order Availability by Location'  
)  
  
fig.update_layout(  
    barmode='stack',  
    font=dict(family='Montserrat'),  
  
    xaxis_title='Restaurant Count',  
    yaxis_title='Location'  
)  
  
fig.show()
```

Visualizing Online Order

```
In [290... fig = px.histogram( df, y='online_order', orientation='h', color='online_order',
                        color_discrete_sequence=[ '#19D3F3', '#636EFA' ])

fig.update_layout(font=dict(family='Montserrat'),

                  title='Online Order Count',
                  legend_title='Online Order',

                  yaxis_title='Online Order',
                  xaxis_title='Count',

                  )

fig.show()
```

Visualizing Book Table

```
In [291... fig = px.histogram( df,y='book_table', color='book_table',
                        color_discrete_map={ 'Yes': '#00E5FF', 'No': '#FFA15A'}
                        )

fig.update_layout( font=dict( family='Montserrat'),

                    title='Table Booking Availability',
                    legend_title='Book Table',

                    xaxis_title='Count',
                    yaxis_title='Book Table'
                    )

fig.show()
```

Visualizing Online Order vs Rate

```
In [292... fig = px.box( df, x='online_order', y='rate', color='online_order',  
    color_discrete_map={ 'Yes': '#00E5FF', 'No': '#FFA15A'}  
)  
fig.update_layout(font=dict(family='Montserrat'),  
    title='Ratings vs Online Order',  
    legend_title='Online Order',  
    xaxis_title='Online Order',  
    yaxis_title='Rating'  
)  
fig.show()
```


Visualizing Book Table vs Rate

```
In [293... fig = px.box( df, x='book_table', y='rate', color='book_table',  
                    color_discrete_map={'Yes': '#00E5FF', 'No': '#FFA15A'})  
  
fig.update_layout(font=dict(family='Montserrat'),  
                  title='Ratings vs Table Booking',  
                  legend_title='Book Table',  
                  xaxis_title='Book Table',  
                  yaxis_title='Rating'  
                  )  
  
fig.show()
```

Visualizing Online Order Facility, Location Wise

```
In [294... df1 = df.groupby(['location', 'online_order'])['name'].count()
df1.to_csv('location_online.csv')
df1 = pd.read_csv('location_online.csv')
df1 = pd.pivot_table(df1, values=None, index=['location'], columns=['online_or
df1
```

Out[294...

online_order	name	
	No	Yes
location		
BTM	1760	3287
Banashankari	397	505
Banaswadi	302	338
Bannerghatta Road	684	918
Basavanagudi	243	439
Bellandur	512	751
Brigade Road	537	655
Brookefield	238	417
Church Street	197	327
Commercial Street	228	142
Cunningham Road	168	307
Domlur	247	233
Ejipura	214	218
Electronic City	676	569
Frazer Town	284	424
HSR	584	1894
Indiranagar	688	1296
JP Nagar	904	1304
Jayanagar	549	1360
Kalyan Nagar	349	488
Kammanahalli	263	375
Koramangala 1st Block	383	841
Koramangala 4th Block	433	557
Koramangala 5th Block	819	1564
Koramangala 6th Block	443	674
Koramangala 7th Block	385	777
Lavelle Road	309	194
MG Road	502	354
Malleshwaram	303	409

	name	
	online_order	No Yes
location		
Marathahalli	689	1088
New BEL Road	253	383
Old Airport Road	216	216
Rajajinagar	284	298
Residency Road	416	247
Richmond Road	548	242
Sarjapur Road	317	711
Shanti Nagar	286	219
Shivajinagar	346	144
St. Marks Road	168	164
Ulsoor	386	614
Whitefield	979	1109
others	2060	2886

```
In [295... plt.style.use('dark_background')

ax = df1.plot(
    kind='bar',
    figsize=(20,12),
    color=['#FFA15A', '#00E5FF']
)

plt.title('Online Order Availability by Location')

plt.xlabel('Location')

plt.ylabel('Restaurant Count')

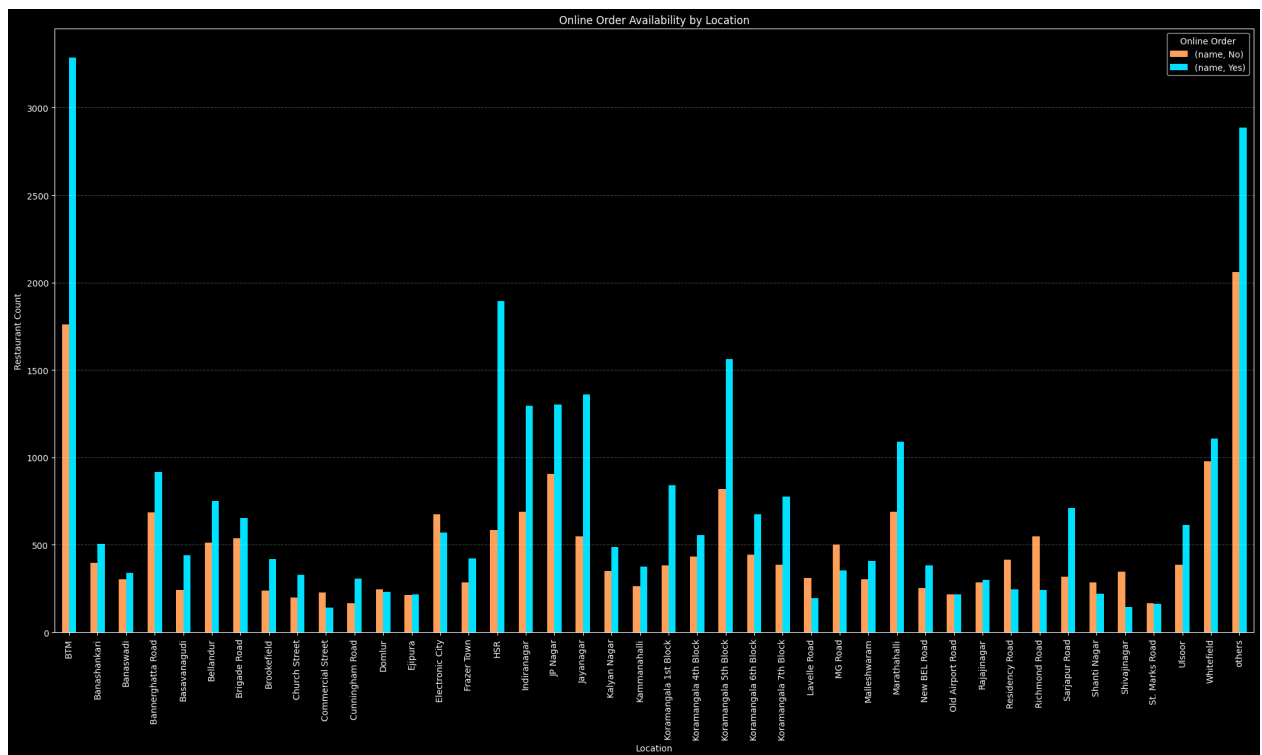
plt.xticks(rotation=90)

plt.legend(title='Online Order')

plt.grid(axis='y', linestyle='--', alpha=0.3)

plt.tight_layout()

plt.show()
```



Visualizing Book Table Facility, Location Wise

```
In [296.. df2 = df.groupby(['location', 'book_table'])['name'].count()
df2.to_csv('location_booktable.csv')
df2 = pd.read_csv('location_booktable.csv')
df2 = pd.pivot_table(df2, values=None, index=['location'], columns=['book_table'])
df2
```

Out[296...

	name	
book_table	No	Yes
location		
BTM	4880	167
Banashankari	839	63
Banaswadi	632	8
Bannerghatta Road	1509	93
Basavanagudi	666	16
Bellandur	1165	98
Brigade Road	1033	159
Brookefield	581	74
Church Street	355	169
Commercial Street	370	0
Cunningham Road	315	160
Domlur	425	55
Ejipura	432	0
Electronic City	1148	97
Frazer Town	701	7
HSR	2268	210
Indiranagar	1548	436
JP Nagar	1897	311
Jayanagar	1632	277
Kalyan Nagar	691	146
Kammanahalli	589	49
Koramangala 1st Block	1175	49
Koramangala 4th Block	677	313
Koramangala 5th Block	1760	623
Koramangala 6th Block	1008	109
Koramangala 7th Block	1004	158
Lavelle Road	282	221
MG Road	539	317
Malleshwaram	630	82

	name	
book_table	No	Yes
location		
Marathahalli	1634	143
New BEL Road	584	52
Old Airport Road	378	54
Rajajinagar	546	36
Residency Road	514	149
Richmond Road	678	112
Sarjapur Road	889	139
Shanti Nagar	451	54
Shivajinagar	467	23
St. Marks Road	218	114
Ulsoor	824	176
Whitefield	1844	244
others	4581	365

```
In [297... plt.style.use('dark_background')

ax = df2.plot(
    kind='bar',
    figsize=(15,8),
    color=['#FFA15A', '#00E5FF']
)

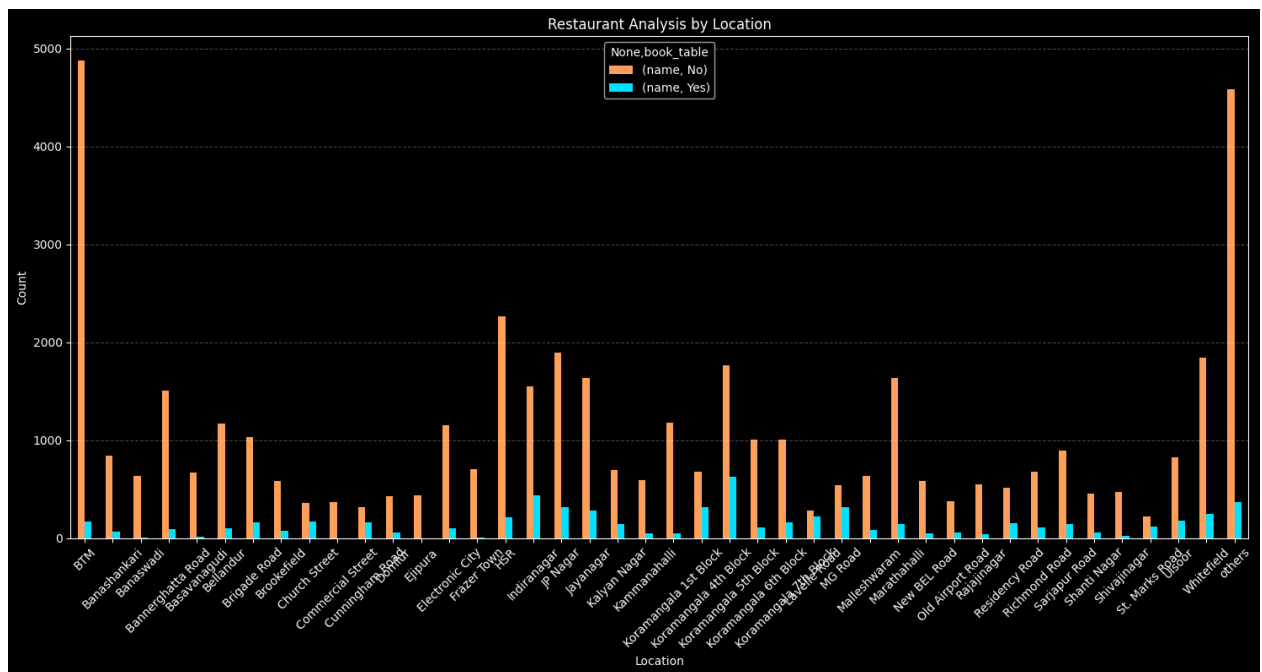
plt.title('Restaurant Analysis by Location')
plt.xlabel('Location')
plt.ylabel('Count')

plt.xticks(rotation=45)

plt.grid(axis='y', linestyle='--', alpha=0.3)

plt.tight_layout()

plt.show()
```



Visualizing Types of Restaurants vs Rate

```
In [298... plt.figure(figsize=(14,8))

sns.boxplot(
    x='Type',
    y='rate',
    data=df,
    palette=['#FFA15A', '#00E5FF']
)

plt.title('Restaurant Type vs Rating')

plt.xlabel('Restaurant Type')

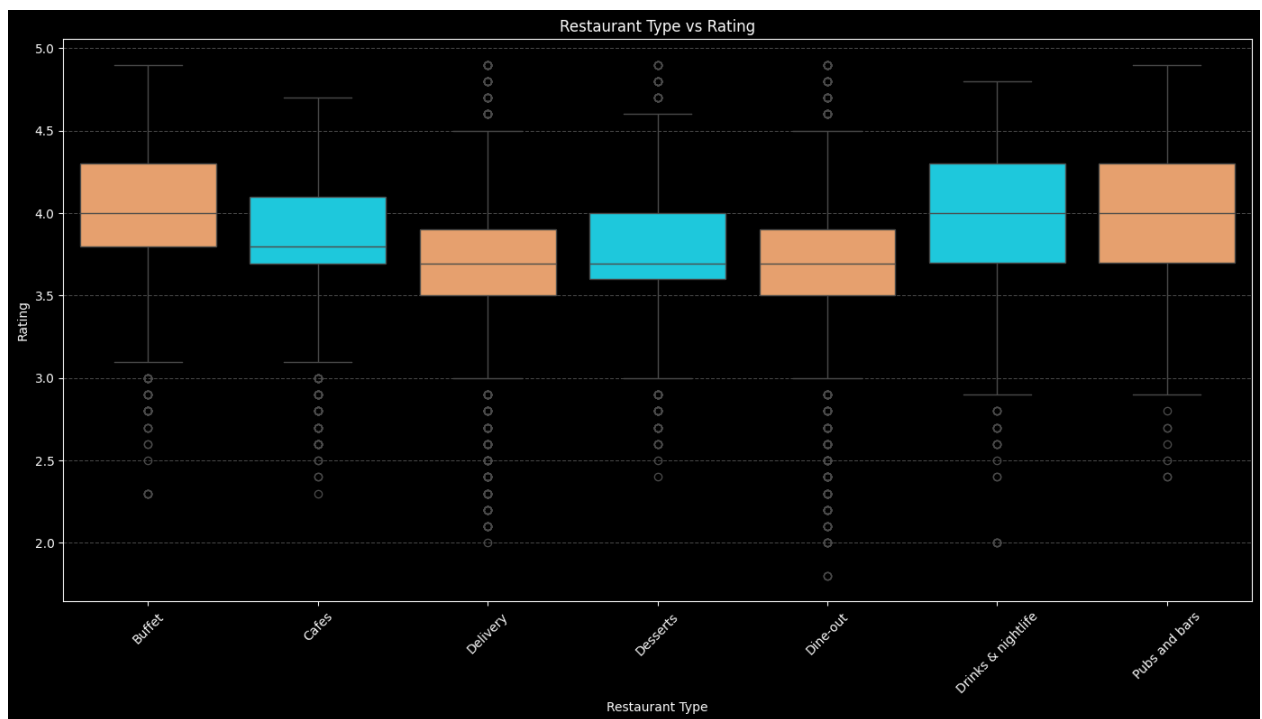
plt.ylabel('Rating')

plt.xticks(rotation=45)

plt.grid(axis='y', linestyle='--', alpha=0.3)

plt.tight_layout()

plt.show()
```

Grouping Types of Restaurants, location wise

```
In [299... df3 = df.groupby(['location', 'Type'])['name'].count()
df3.to_csv('location_Type.csv')
df3 = pd.read_csv('location_Type.csv')
df3 = pd.pivot_table(df3, values=None, index=['location'], columns=['Type'], f
df3
```

Out[299...

location	Type	Buffet	Cafes	Delivery	Desserts	Dine-out	Drinks & nightlife	name Pubs and bars
BTM		21	83	3047	198	1657	22	19
Banashankari		7	36	418	71	356	14	0
Banaswadi		0	24	310	37	262	6	1
Bannerghatta Road		9	46	824	137	576	8	2
Basavanagudi		7	10	344	66	250	5	0
Bellandur		28	36	615	75	476	17	16
Brigade Road		25	46	494	107	448	50	22
Brookefield		6	17	338	45	245	4	0
Church Street		16	45	184	28	194	35	22
Commercial Street		0	13	121	77	159	0	0
Cunningham Road		29	31	192	23	180	14	6
Domlur		15	13	259	35	135	12	11
Ejipura		0	0	245	16	171	0	0
Electronic City		23	24	569	71	516	21	21
Frazer Town		1	11	465	55	172	2	2
HSR		17	47	1687	120	575	14	18
Indiranagar		36	94	1068	139	520	62	65
JP Nagar		44	76	1148	165	718	50	7
Jayanagar		27	77	1039	182	572	12	0
Kalyan Nagar		9	45	366	88	311	18	0
Kammanahalli		2	27	329	35	239	6	0
Koramangala 1st Block		3	24	708	70	396	7	16
Koramangala 4th Block		21	52	458	75	293	60	31
Koramangala 5th Block		59	143	1051	204	804	68	54

								name
	Type	Buffet	Cafes	Delivery	Desserts	Dine-out	Drinks & nightlife	Pubs and bars
location								
Koramangala 6th Block		18	43	507	68	407	51	23
Koramangala 7th Block		25	52	502	127	414	17	25
Lavelle Road		30	27	122	50	183	57	34
MG Road		51	70	260	68	327	46	34
Malleshwaram		11	31	269	85	284	18	14
Marathahalli		30	32	972	104	618	19	2
New BEL Road		4	28	334	33	222	7	8
Old Airport Road		10	5	200	35	163	11	8
Rajajinagar		10	4	254	55	249	3	7
Residency Road		20	31	187	62	285	52	26
Richmond Road		61	18	253	78	352	16	12
Sarjapur Road		22	22	553	82	312	15	22
Shanti Nagar		9	22	198	39	227	8	2
Shivajinagar		6	17	139	37	276	7	8
St. Marks Road		5	9	110	10	139	37	22
Ulsoor		16	55	451	68	357	23	30
Whitefield		28	51	1037	133	763	45	31
others		83	133	2783	276	1549	75	47

```
In [300... plt.style.use('dark_background')

ax = df3.plot(
    kind='bar',
    figsize=(24,10),
    color=['#FFA15A', '#00E5FF']
)

plt.title('Restaurant Analysis')
```

```
plt.xlabel('Category')

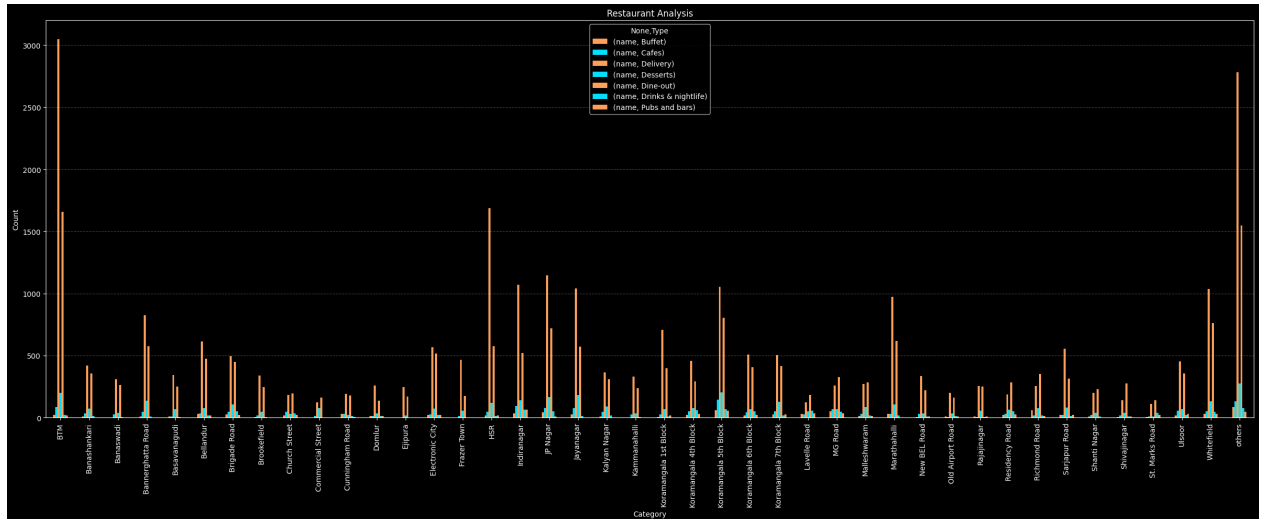
plt.ylabel('Count')

plt.xticks(rotation=90)

plt.grid(axis='y', linestyle='--', alpha=0.3)

plt.tight_layout()

plt.show()
```



No. of Votes, Location Wise

```
In [301... df4 = df[['location', 'votes']]
df4.drop_duplicates()
df5 = df4.groupby(['location'])['votes'].sum()
df5 = df5.to_frame()
df5 = df5.sort_values('votes', ascending=False)
df5.head()
```

```
Out[301...      votes
location
Koramangala 5th Block 1841713
Indiranagar 1086607
Koramangala 4th Block 604544
JP Nagar 580233
BTM 570824
```

```
In [302... plt.figure(figsize=(15,8))
```

```

sns.barplot(
    x=df5.index,
    y=df5['votes'],
    palette=['#FFA15A', '#00E5FF']
)

plt.title('Votes by Restaurant')

plt.xlabel('Restaurant')

plt.ylabel('Votes')

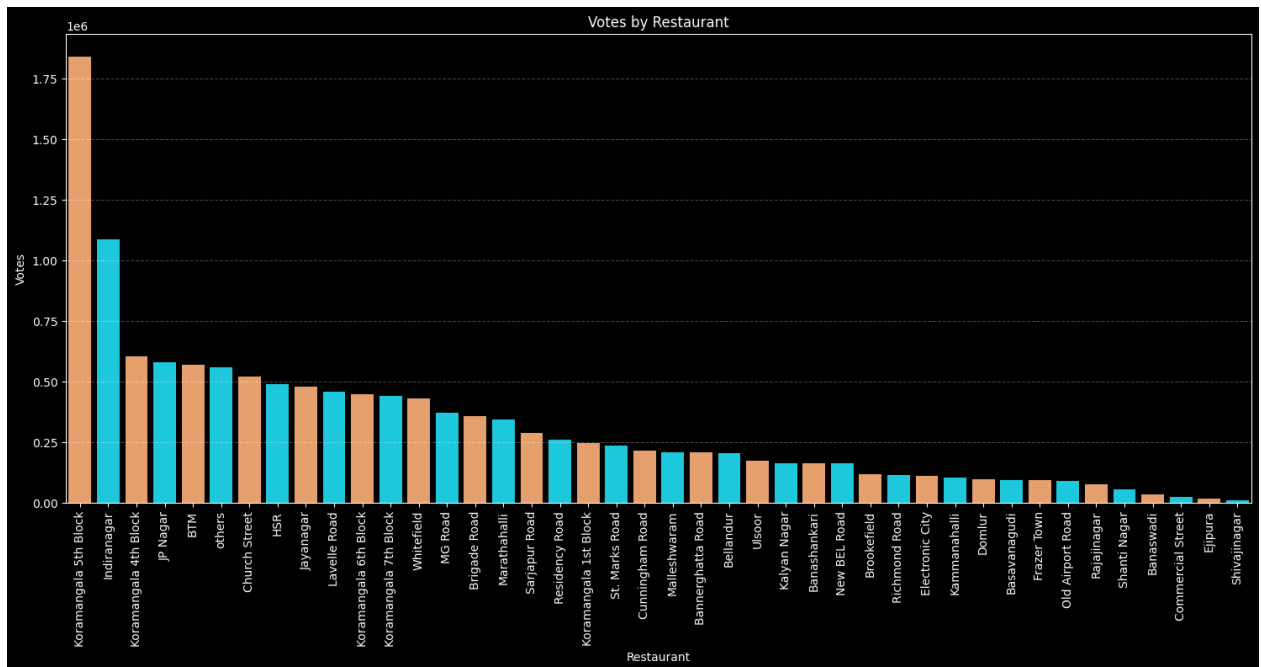
plt.xticks(rotation=90)

plt.grid(axis='y', linestyle='--', alpha=0.3)

plt.tight_layout()

plt.show()

```



In [303... df.head()

Out[303...

	name	online_order	book_table	rate	votes	location	rest_type	cuisi
0	Jalsa	Yes	Yes	4.1	775	Banashankari	Casual Dining	otr
1	Spice Elephant	Yes	No	4.1	787	Banashankari	Casual Dining	otr
2	San Churro Cafe	Yes	No	3.8	918	Banashankari	others	otr
3	Addhuri Udupi Bhojana	No	No	3.7	88	Banashankari	Quick Bites	Sc Ind No Inc
4	Grand Village	No	No	3.8	166	Basavanagudi	Casual Dining	otr

Visualizing Top Cuisines

In [304...

```
df6 = df[['cuisines', 'votes']]
df6.drop_duplicates()
df7 = df6.groupby(['cuisines'])['votes'].sum()
df7 = df7.to_frame()
df7 = df7.sort_values('votes', ascending=False)
df7.head()
```

Out[304...

	votes
cuisines	
others	10266520
North Indian	506442
North Indian, Chinese	254149
South Indian	153031
Chinese	101728

In [305...

```
df7 = df7.iloc[1:, :]
df7.head()
```

Out[305...

votes	
cuisines	
North Indian	506442
North Indian, Chinese	254149
South Indian	153031
Chinese	101728
North Indian, Mughlai	98343

In [306...

```
plt.figure(figsize=(15,8))

sns.barplot(
    x=df7.index,
    y=df7['votes'],
    palette=['#FFA15A', '#00E5FF']
)

plt.title('Votes Analysis')

plt.xlabel('Category')

plt.ylabel('Votes')

plt.xticks(rotation=90)

plt.grid(axis='y', linestyle='--', alpha=0.3)

plt.tight_layout()

plt.show()
```

